Section A

1. (a) gradient of g is -2 (may be seen in function, do not accept $-2x+3$) ***(A1)***

$g(x)=-2x$ ***A1***

[2 marks]

(b) gradient is $\frac{1}{2}$ (may be seen in function) ***(A1)***

attempt to substitute **their** gradient and $(-1,2)$ into any form of equation for straight line ***(M1)***

$y-2=\frac{1}{2}(x+1)$ OR $2=\frac{1}{2}\cdot(-1)+c$

$h(x)=\frac{1}{2}(x+1)+2$ $\left(=\frac{1}{2}x+\frac{5}{2}\right)$ ***A1***

[3 marks]

(c) $(g\circ h)(x)=-2\left(\frac{1}{2}x+\frac{5}{2}\right)$ OR $h(0)=\frac{5}{2}$ OR $g\left(\frac{5}{2}\right)$ ***(A1)***

$(g\circ h)(0)=-5$ ***A1***

[2 marks]

Total [7 marks]

2. $g'(x) = 2x\,e^{x^2+1}$ ***(A2)***

substitute $x = -1$ into **their** derivative ***(M1)***

$g'(-1) = -2\,e^2$ ***A1***

Note: Award ***A0M0A0*** in cases where candidate's incorrect derivative is $g'(x) = e^{x^2+1}$.

[4 marks]

3. (a) (i) attempt to find midpoint of A and B ***(M1)***

centre $\left(-1,3,-2\right)$ (accept vector notation and/or missing brackets) ***A1***

(ii) attempt to find AB or half of AB or distance between the centre and A (or B) ***(M1)***

$$\frac{\sqrt{4^2+2^2+4^2}}{2} \text{ or } \sqrt{2^2+1^2+2^2}$$

$=3$ ***A1***

[4 marks]

(b) attempt to find the distance between their centre and V
(the perpendicular height of the cone) ***(M1)***

$\sqrt{0^2+4^2+2^2}$ OR $\sqrt{\left(\text{their slant height}\right)^2-\left(\text{their radius}\right)^2}$

$=\sqrt{20}\left(=2\sqrt{5}\right)$ ***(A1)***

Volume $=\frac{1}{3}\pi 3^2\sqrt{20}$

$=3\pi\sqrt{20}\left(=6\pi\sqrt{5}\right)$ ***A1***

[3 marks]

Total [7 marks]

4. (a)

Note: Award a maximum of ***M1A0A0*** if the candidate manipulates both sides of the equation (such as moving terms from one side to the other).

METHOD 1 (working with LHS)

attempting to expand $\left(a^2-1\right)^2$ (do not accept a^4+1 or a^4-1) ***(M1)***

$$\text{LHS}=a^2+\frac{a^4-2a^2+1}{4} \text{ or } \frac{4a^2+a^4-2a^2+1}{4}$$ ***A1***

$$=\frac{a^4+2a^2+1}{4}$$ ***A1***

$$=\left(\frac{a^2+1}{2}\right)^2 \ (=\text{RHS})$$ ***AG***

Note: Do not award the final ***A1*** if further working contradicts the AG.

METHOD 2 (working with RHS)

attempting to expand $\left(a^2+1\right)^2$ ***(M1)***

$$\text{RHS}=\frac{a^4+2a^2+1}{4}$$

$$=\frac{4a^2+a^4-2a^2+1}{4}$$ ***A1***

$$=a^2+\frac{a^4-2a^2+1}{4}$$ ***A1***

$$=a^2+\left(\frac{a^2-1}{2}\right)^2 \ (=\text{LHS})$$ ***AG***

Note: Do not award the final ***A1*** if further working contradicts the ***AG***.

[3 marks]

continued…

Question 4 continued

(b) recognise base and height as a and $\left(\dfrac{a^2-1}{2}\right)$ (may be seen in diagram) ***(M1)***

correct substitution into triangle area formula ***A1***

$$\text{Area} = \frac{a}{2}\left(\frac{a^2-1}{2}\right) \text{ (or equivalent) } \left(= \frac{a\left(a^2-1\right)}{4} = \frac{a^3-a}{4}\right)$$

[2 marks]

Total [5 marks]

5. (a) $f'(4) = 6$ ***A1***

[1 mark]

(b) $f(4) = 6 \times 4 - 1 = 23$ ***A1***

[1 mark]

(c) $h(4) = f\left(g(4)\right)$ ***(M1)***

$h(4) = f(4^2 - 3 \times 4) = f(4)$

$h(4) = 23$ ***A1***

[2 marks]

(d) attempt to use chain rule to find h' ***(M1)***

$f'\left(g\left(x\right)\right) \times g'\left(x\right)$ OR $\left(x^2 - 3x\right)' \times f'\left(x^2 - 3x\right)$

$h'(4) = \left(2 \times 4 - 3\right) f'\left(4^2 - 3 \times 4\right)$ ***A1***

$= 30$

$y - 23 = 30\left(x - 4\right)$ OR $y = 30x - 97$ ***A1***

[3 marks]

Total [7 marks]

6. (a) $\mathrm{P}(A \cap B) = 0.24$ ***A1***

[1 mark]

(b) $\mathrm{P}(A \cup B) = 1.1 - \mathrm{P}(A \cap B)$ ***(A1)***

$(0 \le)\mathrm{P}(A \cup B) \le 1$ ***(M1)***

Note: This may be conveyed in a clearly labelled diagram or written explanation where $\mathrm{P}(A \cup B) = 1$

the minimum value of $\mathrm{P}(A \cap B)$ is 0.1 ***A1***

[3 marks]

(c) A is a subset of B (so $\mathrm{P}(A \cap B) = \mathrm{P}(A)$). ***R1***

Note: This may be conveyed in a clearly labelled diagram where A is completely inside B, or in a written explanation indicating that $\mathrm{P}(A \cap B) = \mathrm{P}(A)$

so the maximum value of $\mathrm{P}(A \cap B)$ is 0.3 ***A1***

Note: Do not award ***R0A1***.

[2 marks]

Total [6 marks]

Section B

7. (a) correct substitution of $h=3$ and $k=2$ into $f(x)$ ***(A1)***

$f(x)=a(x-3)^2+2$

correct substitution of $(5,0)$ ***(A1)***

$0=a(5-3)^2+2\ \left(a=-\frac{1}{2}\right)$

Note: The first two A marks are independent.

$f(x)=-\frac{1}{2}(x-3)^2+2$ ***A1***

[3 marks]

(b) (i) **METHOD 1**

correct substitution of $(1,\ 4)$ ***(A1)***

$p+(t-1)-p=4$

$t=5$ ***A1***

substituting their value of t into $9p-3(t-1)-p=4$ ***(M1)***

$8p-12=4$

$p=2$ ***A1***

METHOD 2

correct substitution of ONE of the coordinates $(-3,4)$ or $(1,4)$ ***(A1)***

$9p-3(t-1)-p=4$ OR $p+(t-1)-p=4$

valid attempt to solve their two equations ***(M1)***

$p=2, t=5$ ***A1A1***

$\left(g(x)=2x^2+4x-2\right)$

(ii) attempt to find the x-coordinate of the vertex ***(M1)***

$x=\frac{-3+1}{2}(=-1)$ OR $\frac{-4}{2\times 2}$ OR $4x+4=0$ OR $2(x+1)^2-4$

y-coordinate of the vertex $=-4$ ***(A1)***

correct range ***A1***

$[-4,+\infty[$ OR $y\geq -4$ OR $g\geq -4$ OR $[-4,\infty)$

[7 marks]

(c) equating the two functions or equations ***(M1)***

$g(x) = j(x)$ OR $px^2 + (t-1)x - p = -x + 3p$

$px^2 + tx - 4p = 0$ ***(A1)***

attempt to find discriminant (do not accept only in quadratic formula) ***(M1)***

$\Delta = t^2 + 16p^2$ ***A1***

$\Delta = t^2 + 16p^2 > 0$, because $t^2 \geq 0$ <u>and</u> $p^2 > 0$, therefore the sum will be positive ***R1R1***

Note: Award ***R1*** for recognising that Δ is positive and ***R1*** for the reason.

There are two distinct points of intersection between the graphs of g and j. ***AG***

[6 marks]

Total [16 marks]

8. (a) (i) valid approach to find the required logarithm ***(M1)***

$2^x = \frac{1}{16}$ OR $2^x = 2^{-4}$ OR $\frac{1}{16} = 2^{-4}$ OR $\log_2 1 - \log_2 16$

$\log_2 \frac{1}{16} = -4$ ***A1***

(ii) valid approach to find the required logarithm ***(M1)***

$9^x = 3$ OR $3^{2x} = 3$ OR $3 = 9^{\frac{1}{2}}$ OR $\frac{\log_3 3}{\log_3 9}$

$\log_9 3 = \frac{1}{2}$ ***A1***

(iii) $\left(\sqrt{3}\right)^x = 81$ OR $\frac{\log_3 81}{\log_3 \sqrt{3}}$ ***(A1)***

$(3)^{\frac{x}{2}} = 3^4$ OR $\frac{x}{2} = 4$ OR $\frac{4}{\frac{1}{2}}$ ***(A1)***

$x = 8$ ***A1***

[7 marks]

continued…

Question 8 continued

(b) (i)

Note: There are many valid approaches to the question, and the steps may be seen in different ways. Some possible methods are given here, but candidates may use a combination of one or more of these methods. In all methods, the final ***A*** mark is awarded for working which leads directly to the ***AG***.

METHOD 1

$(ab)^3 = a$ ***(A1)***

attempt to isolate b or a power of b ***(M1)***

correct working ***(A1)***

$b = \frac{a}{a^3b^2}$ OR $b^3 = a^{-2}$ OR $b^{-1} = (ab)^2$ OR $b^3 = \frac{1}{a^2}$

$b = \frac{1}{a^2b^2}$ OR $b = (ab)^{-2}$ OR $3\log_{ab} b = -2\log_{ab} a$ OR $-\log_{ab} b = 2\log_{ab} ab$ ***A1***

$\log_{ab} b = -2$ ***AG***

METHOD 2

$(ab)^3 = a$ ***(A1)***

taking logarithm to base ab on both sides ***(M1)***

$\log_{ab}(ab)^3 = \log_{ab} a$ OR $\log_{ab} a^3 b^3 = \log_{ab} a$

correct application of log rules leading to equation in terms of $\log_{ab}$ ***(A1)***

$3\log_{ab} a + 3\log_{ab} b = \log_{ab} a$ OR $3\log_{ab} b = -2\log_{ab} a$ OR $\log_{ab} b^3 = \log_{ab} a^{-2}$

$\log_{ab} b = \log_{ab} a^{-\frac{2}{3}}$ OR $\log_{ab} b = -\frac{2}{3}\log_{ab} a$ OR $\log_{ab} b = -\frac{2}{3}(3)$ ***A1***

$\log_{ab} b = -2$ ***AG***

Note: Candidates may substitute $\log_{ab} a = 3$ at any point in their working.

continued…

Question 8 continued

METHOD 3

$\log_{ab} a = 3$

writing in terms of base a ***(M1)***

$\dfrac{\log_a a}{\log_a ab}(=3)$

correct application of log rules ***(A1)***

$\dfrac{\log_a a}{\log_a a + \log_a b}(=3)$ OR $\dfrac{1}{1+\log_a b}(=3)$ OR $3\log_a b = -2$ OR

$\log_a b = -\dfrac{2}{3}$

writing $\log_{ab} b$ in terms of base a ***(A1)***

$\log_{ab} b = \dfrac{\log_a b}{\log_a a + \log_a b}$

correct working ***A1***

$\log_{ab} b = \dfrac{-\frac{2}{3}}{1-\frac{2}{3}}$ OR $\dfrac{\left(-\frac{2}{3}\right)}{\left(\frac{1}{3}\right)}$

$\log_{ab} b = -2$ ***AG***

continued…

Question 8 continued

METHOD 4

$\log_{ab} ab = 1$ ***A2***

$\log_{ab} a + \log_{ab} b = 1$ ***(A1)***

$3 + \log_{ab} b = 1$ ***A1***

$\log_{ab} b = -2$ ***AG***

(ii) applying the quotient rule or product rule for logs

$\log_{ab} \frac{\sqrt[3]{a}}{\sqrt{b}} = \log_{ab} \sqrt[3]{a} - \log_{ab} \sqrt{b}$ OR $\log_{ab} \frac{\sqrt[3]{a}}{\sqrt{b}} = \log_{ab} \sqrt[3]{a} + \log_{ab} \frac{1}{\sqrt{b}}$ ***(A1)***

correct working ***(A1)***

$= \frac{1}{3}\log_{ab} a - \frac{1}{2}\log_{ab} b$ OR $\log_{ab} ab - \log_{ab} \sqrt{b}$

$= \frac{1}{3} \cdot 3 - \frac{1}{2}(-2)$ ***(A1)***

$= 2$ ***A1***

Note: Award ***A1A0A0A1*** for a correct answer with no working.

[8 marks]

Total [15 marks]

9. (a) $\cos^2 x - 3\sin^2 x = 0$

valid attempt to reduce equation to one involving one trigonometric function ***(M1)***

$\frac{\sin^2 x}{\cos^2 x} = \frac{1}{3}$ OR $1 - \sin^2 x - 3\sin^2 x = 0$ OR $\cos^2 x - 3\left(1 - \cos^2 x\right) = 0$

OR $\cos 2x - 1 + \cos 2x = 0$

correct equation ***(A1)***

$\tan^2 x = \frac{1}{3}$ OR $\cos^2 x = \frac{3}{4}$ OR $\sin^2 x = \frac{1}{4}$ OR $\cos 2x = \frac{1}{2}$

$\tan x = \pm\frac{1}{\sqrt{3}}$ OR $\cos x = \pm\frac{\sqrt{3}}{2}$ OR $\sin x = (\pm)\frac{1}{2}$ OR $2x = \frac{\pi}{3}\left(, \frac{5\pi}{3}\right)$ ***(A1)***

$x = \frac{\pi}{6}, \ x = \frac{5\pi}{6}$ ***A1A1***

Note: Award ***M1A1A0A1A0*** for candidates who omit the $\pm$ (for tan or cos) and give only $x = \frac{\pi}{6}$.

Award ***M1A1A0A0A0*** for candidates who omit the $\pm$ (for tan or cos) and give only $x = 30°$.

Award ***M1A1A1A1A0*** for candidates who give both answers in degrees.

Award ***M1A1A1A1A0*** for candidates who give both correct answers in radians, but who include additional solutions outside the domain.

Award a maximum of ***M1A0A0A1A1*** for correct answers with no working.

[5 marks]

continued…

Question 9 continued

(b) (i) attempt to use the chain rule (may be evidenced by at least one $\cos x \sin x$ term) ***(M1)***

$f'(x) = -2\cos x \sin x - 6\sin x \cos x \left(= -8\sin x \cos x = -4\sin 2x\right)$ ***A1***

(ii) valid attempt to solve their $f'(x) = 0$ ***(M1)***

At least 2 correct x-coordinates (may be seen in coordinates) ***(A1)***

$x = 0$, $x = \frac{\pi}{2}$, $x = \pi$

Note:	Accept additional correct solutions outside the domain.
	Award ***A0*** if any additional incorrect solutions are given.

correct coordinates (may be seen in graph for part (c)) ***A1A1A1***

$(0,1)$, $(\pi,1)$, $\left(\frac{\pi}{2},-3\right)$

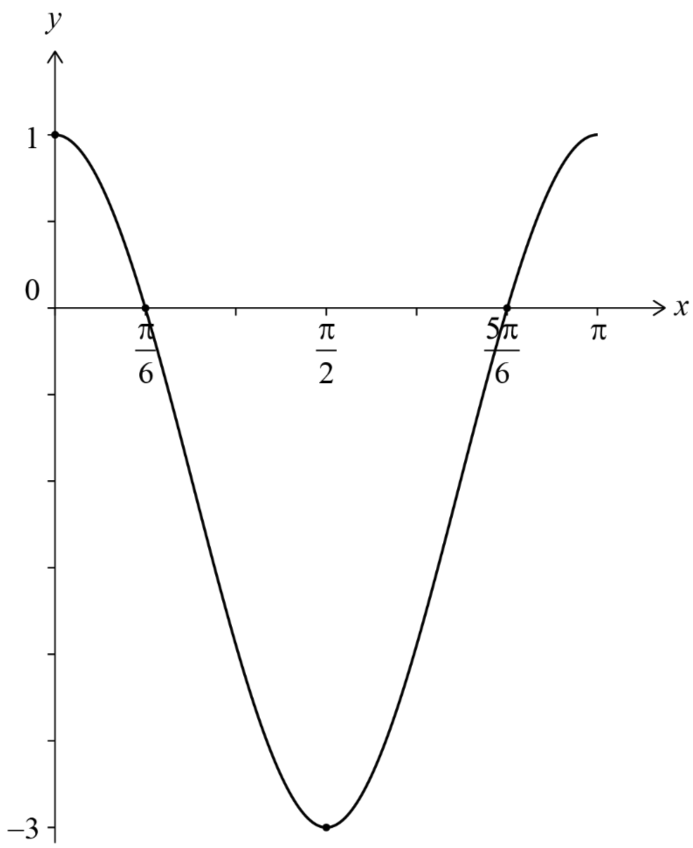
Note:	Award a maximum of ***M1A1A1A1A0*** if any additional solutions are given.
Note:	If candidates do not find at least two correct x-coordinates, it is possible to award the appropriate final marks for their correct coordinates, such as ***M1A0A0A1A0.***

[7 marks]

continued…

Question 9 continued

(c)



Note: In this question do not award follow through from incorrect values found in earlier parts.

approximately correct smooth curve with minimum at $\left(\frac{\pi}{2}, -3\right)$ ***A1***

Note: If candidates do not gain this mark then award no further marks.

endpoints at $(0,1)$, $(\pi,1)$, x-intercepts at $\frac{\pi}{6}$, $\frac{5\pi}{6}$ ***A1***

correct concavity clearly shown at $(0,1)$ and $(\pi,1)$ ***A1***

Note: The final two marks may be awarded independently of each other.

[3 marks]

Total [15 marks]